DECISION MAKING IN C

Decision making in **C programming** allows the program to **make logical choices** and **execute specific code blocks** based on certain conditions. This is an essential concept in controlling the **flow of execution** in a program.

🧠 1. What is Decision Making in C?

Decision making in C refers to executing different code segments based on whether a **condition** is **true or false**. C provides several control structures for this purpose.

✅ 2. Common Decision-Making Statements in C

Statement	Description
if	Executes a block if the condition is true
if...else	Executes one block if true, another if false
else if ladder	Checks multiple conditions in sequence
switch	Executes code based on matching cases
Ternary Operator	Shortened form of if...else

🔹 3. The if Statement

```
if (condition) {
  // Code to execute if condition is true
}
```

Example:

```
int x = 10;
if (x > 5) {
  printf("x is greater than 5\n");
}
```

🔸 4. The if...else Statement

```
if (condition) {
   // Code if condition is true
} else {
   // Code if condition is false
}
```

Example:

```
int x = 3;
if (x > 5) {
   printf("x is greater than 5\n");
} else {
   printf("x is not greater than 5\n");
}
```

🔺 5. The else if Ladder

```
if (condition1) {
   // Block 1
} else if (condition2) {
   // Block 2
} else {
   // Default Block
}
```

Example:

```
int x = 0;
if (x > 0) {
   printf("Positive\n");
} else if (x < 0) {
   printf("Negative\n");
} else {
```

```
    printf("Zero\n");
}
```

🧩 6. The switch Statement

Used for multiple constant values:

```
switch (expression) {
    case value1:
        // Block 1
        break;
    case value2:
        // Block 2
        break;
    default:
        // Default block
}
```

Example:

```
int day = 3;
switch (day) {
    case 1: printf("Monday\n"); break;
    case 2: printf("Tuesday\n"); break;
    case 3: printf("Wednesday\n"); break;
    default: printf("Invalid day\n");
}
```

🎯 7. The Ternary Operator (?:)

A shorthand for if...else.

```
(condition) ? expression_if_true : expression_if_false;
```

Example:

```
int x = 10, y = 5;
int max = (x > y) ? x : y;
printf("Max is %d\n", max);
```

⚙️ 8. Nested Decision Making

You can nest if, if-else, or switch statements.

Example:

```
int x = 10, y = 20;
if (x > 5) {
   if (y > 15) {
      printf("Both conditions are true\n");
   }
}
```